

GIT AND GITHUB PRACTICAL REPORT

1. Git Clone

Command:

```
git clone
```

Description:

The `git clone` command was used to copy the File Management System repository from GitHub to the local computer. This created a local copy of the repository containing all the project files and Git history.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System> git clone https://github.com/Peash02/File-Management-System
Cloning into 'File-Management-System'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

Output of git clone

2. Git Repository Initialization

Command:

```
git init
```

Description:

The `git init` command was used to initialize a directory as a Git repository. This created a `.git` folder, which is used by Git to track changes and maintain the repository history.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System> git init
Initialized empty Git repository in C:/Users/BCA 7/Documents/File_Management_System/.git/
```

Output of git init

3. Checking Status After Cloning

Command:

```
git status
```

Description:

After moving into the cloned `File-Management-System` folder, `git status` was run to confirm the starting state. It shows the repository is on the `main` branch, is up to date with `origin/main`, and has a clean working tree with nothing to commit.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
```

git status on the main branch with a clean working tree

4. Verifying the Remote Repository

Command:

```
git remote -v
```

Description:

The `git remote -v` command lists the remote repositories linked to the local repository. It confirms that `origin` points to the GitHub repository for both fetch and push operations.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git remote -v
origin  https://github.com/Peash02/File-Management-System (fetch)
origin  https://github.com/Peash02/File-Management-System (push)
```

Output of git remote -v

5. Creating a New Branch

Command:

```
git switch -c file-manager
```

Description:

A new branch named `file-manager` was created using the `git switch -c` command. The command also switched the working directory to the newly created branch.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git switch -c file-manager
Switched to a new branch 'file-manager'
```

Creating and switching to the file-manager branch

6. Checking Branches

Command:

```
git branch
```

Description:

The `git branch` command was used to display the branches available in the local repository. The `*` symbol indicates the branch that is currently active, which is `file-manager`.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git branch
* file-manager
  main
```

Output of git branch

7. Checking Repository Status

Command:

```
git status
```

Description:

The `git status` command was used to check the current state of the working directory. After `README.md` was edited on the `file-manager` branch, Git reported it as modified but not yet staged for commit.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git status
On branch file-manager
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
```

git status showing README.md modified but not staged

8. Adding Files to the Staging Area

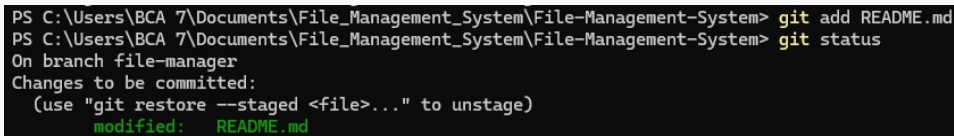
Command:

```
git add README.md git status
```

Description:

The `git add` command was used to add the modified `README.md` file to the staging area, preparing it to be included in the next commit. Running `git status` again shows the file now listed under "Changes to be committed". (`git add .` can be used to stage all changes at once.)

Screenshot:



```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git add README.md
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git status
On branch file-manager
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

Staging README.md and verifying with git status

9. Configuring Git User Details

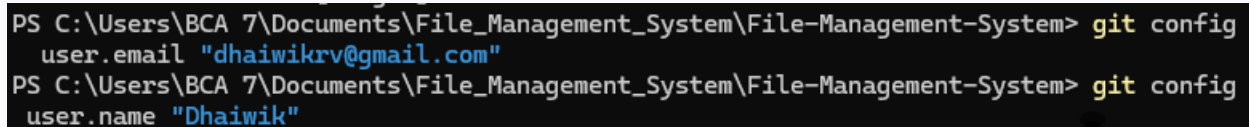
Command:

```
git config user.email "" git config user.name "Dhaiwik"
```

Description:

The `git config` command was used to set the user email and name that Git attaches to every commit. This identifies the author of the changes in the repository history.

Screenshot:



```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git config user.email "dhaiwikrv@gmail.com"
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git config user.name "Dhaiwik"
```

Setting user.email and user.name

10. Committing Changes

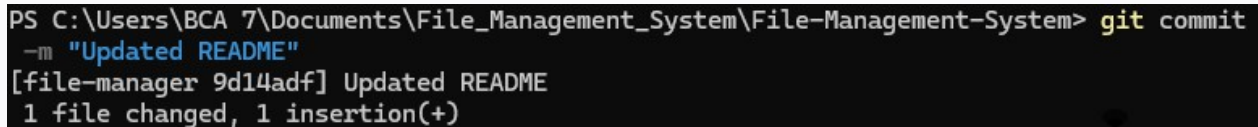
Command:

```
git commit -m "Updated README"
```

Description:

The `git commit` command was used to save the staged changes to the local Git repository. A descriptive commit message was added to identify the changes. Git created commit `9d14adf` on the `file-manager` branch.

Screenshot:



```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git commit
-m "Updated README"
[file-manager 9d14adf] Updated README
1 file changed, 1 insertion(+)
```

Output of git commit

11. Pushing Changes to GitHub

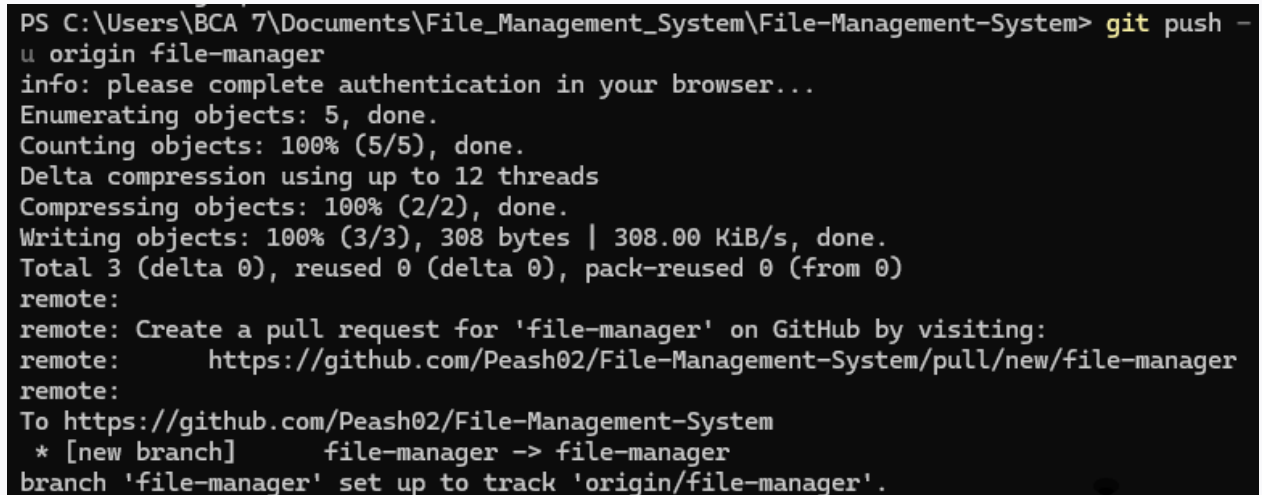
Command:

```
git push -u origin file-manager
```

Description:

The `git push` command was used to upload the local commits from the `file-manager` branch to the remote GitHub repository. The `-u` option sets the remote branch as the upstream branch. GitHub also returned a link to create a pull request for the new branch.

Screenshot:



```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git push -
u origin file-manager
info: please complete authentication in your browser...
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 308 bytes | 308.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'file-manager' on GitHub by visiting:
remote:   https://github.com/Peash02/File-Management-System/pull/new/file-manager
remote:
To https://github.com/Peash02/File-Management-System
 * [new branch]      file-manager -> file-manager
branch 'file-manager' set up to track 'origin/file-manager'.
```

Pushing the file-manager branch to GitHub

12. Switching Between Branches

Command:

```
git switch main
```

Description:

The `git switch` command was used to move from one branch to another. In this case, the working branch was changed from `file-manager` to the `main` branch.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git switch
main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git pull o
rigin main
From https://github.com/Peash02/File-Management-System
* branch          main          -> FETCH_HEAD
Already up to date.
```

Switching to main (followed by git pull origin main)

13. Pulling Changes from GitHub

Command:

```
git pull origin main
```

Description:

The `git pull` command was used to download the latest changes from the GitHub repository and update the local `main` branch. Git reported that the branch was already up to date. This is the second command in the same screenshot as the previous section.

Screenshot:

```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git switch
main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git pull o
rigin main
From https://github.com/Peash02/File-Management-System
* branch          main          -> FETCH_HEAD
Already up to date.
```

git pull origin main after switching to main

14. Merging Branches

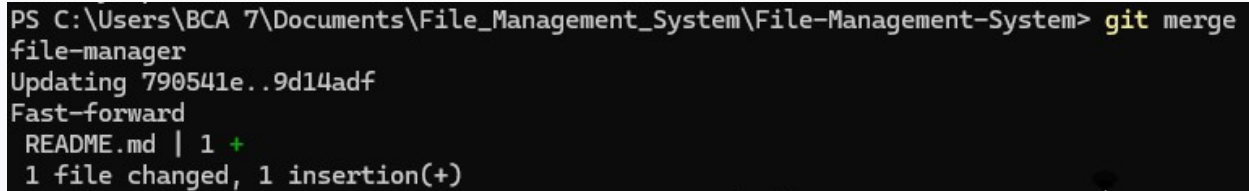
Command:

```
git merge file-manager
```

Description:

The `git merge` command was used to combine the changes from the `file-manager` branch into the `main` branch. Because `main` had no new commits of its own, Git performed a fast-forward merge, bringing the README update into `main`.

Screenshot:



```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git merge  
file-manager  
Updating 790541e..9d14adf  
Fast-forward  
 README.md | 1 +  
 1 file changed, 1 insertion(+)
```

Fast-forward merge of file-manager into main

15. Pushing the Merged Main Branch to GitHub

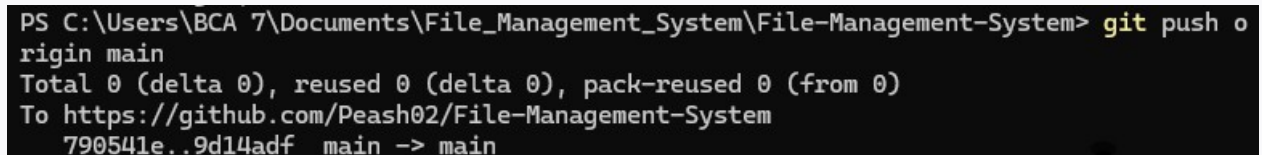
Command:

```
git push origin main
```

Description:

After the merge, `git push origin main` was used to upload the updated `main` branch to GitHub, moving the remote `main` from commit `790541e` to `9d14adf`.

Screenshot:



```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git push o  
rigin main  
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
To https://github.com/Peash02/File-Management-System  
 790541e..9d14adf  main -> main
```

Pushing the merged main branch

16. Viewing Commit History

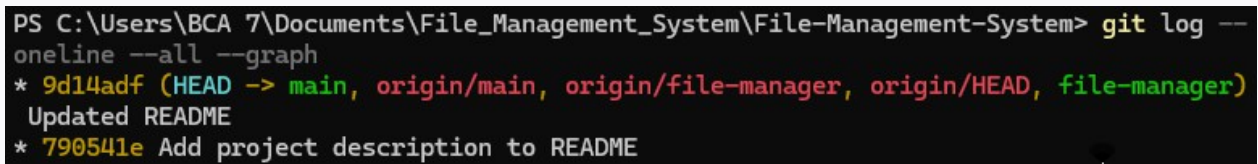
Command:

```
git log --oneline --all --graph
```

Description:

The `git log` command was used to view the commit history of the repository. The `--oneline` option displays commits in a compact format, `--all` includes every branch, and `--graph` shows the branch and merge history visually. Both `main` and `file-manager` (local and remote) point to the same latest commit.

Screenshot:



```
PS C:\Users\BCA 7\Documents\File_Management_System\File-Management-System> git log --oneline --all --graph
* 9d14adf (HEAD -> main, origin/main, origin/file-manager, origin/HEAD, file-manager)
  Updated README
* 790541e Add project description to README
```

Commit history with branch graph